CYRILLE FAVREAU

Scientific visualization architect / Science communicator

cyrille.favreau@gmail.com
+41 78 879 04 76 +33 60 320 26 43
34 rue de la libération, Gaillard (France)
www.linkedin.com/in/cyrillefavreau
github.com/favreau
www.scientific-visualization.org

I am a dedicated professional advancing science and technology, specializing in the design and implementation of high-performance systems for scientific visualization. Recognized as an excellent communicator, I simplify complex concepts. As an advocate for the fusion of art and science, I contribute to innovative projects, producing high-quality media that showcases the synergy between technology, creative expression, and science.

EXPERIENCE

Scientific visualization architect, Science communication

EPFL - Blue Brain Project

2014-Ongoing Geneva, Switzerland

- Responsible for the architecture, design, implementation, and comprehensive documentation of visualization systems, ensuring robust performance and scalability.
- Responsible for leading major presentations of the Blue Brain Project to the general public and VIPs.
- Introduced AI at all levels of the development process: Science understanding, code writing, code reviews, image quality enhancement, and GPT assistant to use the software.
- Creator, architect, and visionary behind Brayns, the official visualization platform for the Blue Brain Project, recognized as a strategic project.
- Developed the Blue Brain BioExplorer, an enhancement of Brayns that broadens visualization capabilities to various scientific fields, including particle physics, virology, nuclear fusion, molecular systems, and more.
- Designed and implemented high-performance algorithms on heterogeneous architectures (CPU and GPU), for visualizing large-scale models on giant screens, bringing interactive ray-tracing to computational neuroscience.
- Key figure in major exhibitions such as Centre Pompidou, EPFL, and CCCB.
- Contributed to over 20 scientific publications in the field of computational neuroscience.
- Featured in prominent publications including Harvard Brain Magazine, NVIDIA Spotlight, and Intel showcase.
- Acknowledged for exceptional science communication and popularization efforts, instrumental in making complex scientific concepts accessible.
- Played a pivotal role in collaborations with CERN and EPFL's experimental museology, driving innovative visualization projects.
- Produced an educational film on the Glucose-COVID19 hypothesis, contributing to public understanding and scientific discourse.

Technical Architect - Vice President

Lombard Odier Darier Hentsch & Cie

2004-2014 Geneva, Switzerland

- Team lead of native libraries in the technical architecture team.
- Lecturer, effectively communicating and explaining new technical solutions.
- Ensured interoperability across various technology stacks.
- Designed low-level libraries and integrated new packages.
- Provided support for development, finding pragmatic solutions to complex problems.
- Recognized as an expert in multi-core optimization and high-performance distributed computing, particularly on heterogeneous architectures.
- Delivered solutions and support tailored to the specific needs of the client.

MOST PROUD OF

Achievement
Created a successful process from the ground up for producing high-quality and high-fidelity media and interactive experiences in computational neuroscience.

Diversity of experiences
From the music industry in London to telecommunications in France, banking and finance in Geneva, and scientific visualization at EPFL.

Health
Weekly swimming marathon of 15 to 20 km

STRENGTHS

- Technical expertise and communication skills
- Energy and Enthusiasm
- Focus and engagement. I deliver!

SIDE PROJECTS

Quantum of Sol-R

Creator

2012-2023

- A sophisticated real-time graphics engine utilizing CUDA/OpenCL, leveraging principles of quantum physics.

Montreux Jazz Festival

Drummer

2013 Montreux

- Performed with the Canettes Blues Band.

EXPERIENCE

Team leader and technical architect for SFR

SOPRA Group

2001-2004 | Rennes, France

- As the team leader and technical architect for SFR:
- Spearheaded C++, UNIX, and WebSphere projects.
- Provided leadership and development support.
- Demonstrated strong expertise in C++ and a deep understanding of HP-UX and Solaris platforms.
- Extensively utilized Oracle databases to drive project success.

Developer Support Engineer

eXcelon Corporation

2000-2001 | Theale, United-Kingdom

- Development support to distinguished clientele, including prominent African and European companies like AXA, SwissRe, Rhenus, and others.
- Demonstrated expertise across a wide spectrum of technologies, including C++, Visual Basic, Java, XML, and XSLT.
- Worked proficiently across various platforms, including Windows, Linux, and UNIX.
- Maintained a customer-oriented focus, delivering tailored solutions to precisely meet the unique needs of each client.
- Ensured projects were successful, efficient, and characterized by high client satisfaction.

Software Developer

Counterpoint Systems Limited

1998-2000 | Camden Town, United-Kingdom

- Maintenance of a robust royalty management system for the music industry.
- Proficiently utilized Delphi and Oracle technologies.
- Contributed to the creation of a system that played a pivotal role in tracking and managing royalties within the dynamic and intricate landscape of the music business.

Software Developer

Thalassa Ingenierie Informatique

1996-1998 | Nantes, France

- Developed and maintained an accounting system at Thalassa Ingénierie Informatique.
- Adeptly employed Delphi and Oracle technologies to ensure seamless and up-to-date software operation.

Soldier

Armée de terre Française

1995-1996 | Angoulème, France

- Military service at 1er Régiment d'Infanterie de Marine

Support Engineer Trainee

Microsoft France

1995 | Les Ulys, France

- Provided expert assistance and guidance to users of Microsoft Access.

EDUCATION

Bachelor of Science

Computer Science

1993-1995 | Wrexham, Wales

Diplome Universitaire Technologie

Informatique

1993-1995 | Nantes, France

ITIL v3 Qualification

Service management

2012 | Geneva, Switerland

LANGUAGES

French

English

Spanish

REFEREES

Dr. Felix Schürmann
@ Director of the Computing Division and Co-Director of the EPFL Blue Brain Project, Senior HPC technologist at Google, Adjunct Professor at EPFL
felix.schuermann@epfl.ch

Dr. René Brun
@ Senior Physicist, Honorary member at CERN
Rene.Brun@cern.ch

Dr. Emmanuelle Logette
@ Research Scientist at EPFL
emanuelle.logette@epfl.ch

IT Team Lead/Trading and Operation
@ Lombard Odier Groups
m.sartor@lombardodier.com

SKILLS

Science communication
System design | Parallel architectures
Scientific visualization | C++ | CUDA
GPGPU | High-performance algorithms
Project management | Tech lead
Ray-tracing | Virtual Reality | Unity